



**CS-103**

# **INFORMATION TECHNOLOGY FOR BUSINESS**

PROJECT REPORT

**Prepared By**

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## 1. Project Objective

The primary objective of this project is to apply the data analytics skills acquired in CS-103—specifically Power BI, Power Query, Data Modeling, and SQL—to a real-world business scenario. Using the publicly available "Superstore Sales Dataset", this project aims to conduct a comprehensive analysis of the store's operations to uncover key business insights.

The analysis will focus on:

- Performance Analysis: Evaluating overall sales, profitability, and discount trends.
- Product Insights: Identifying top-performing and underperforming product categories and sub-categories.
- Geographic & Customer Analysis: Analyzing sales and profit distribution across different regions, states, and customer segments to identify high-value markets.
- Operational Efficiency: Assessing the impact of different shipping modes on business metrics.

The final outcome is an interactive Power BI dashboard that enables stakeholders to explore these insights dynamically, supported by a detailed project report and a set of SQL queries that demonstrate the underlying data manipulation and validation logic. This project serves as a practical application of the entire data analysis workflow, from raw data extraction to the final presentation of actionable insights.

## 2. Data Source Description

The dataset used for this project is the Sample - Superstore dataset, a widely-used resource for retail analytics and business intelligence demonstrations.

- Source: Kaggle (uploaded by user Vivek468).
- URL: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>
- Format: Excel (.xlsx)
- Data Volume: The dataset contains 9,994 rows and 21 columns. It records sales transactions of a superstore over a four-year period (approximately 2014-2017), covering a wide range of retail operations.

The dataset includes detailed information about orders, customers, products, and financial metrics. The table below provides a description of each field.

| Column Name   | Description                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------|
| Row ID        | A unique identifier for each row or record in the dataset.                                                     |
| Order ID      | A unique identifier for each customer order. An order can consist of multiple products, each with its own row. |
| Order Date    | The date on which the order was placed by the customer.                                                        |
| Ship Date     | The date on which the order was shipped to the customer.                                                       |
| Ship Mode     | The shipping method selected by the customer (e.g., Second Class Standard Class, First Class, Same Day).       |
| Customer ID   | A unique identifier for each customer.                                                                         |
| Customer Name | The full name of the customer.                                                                                 |
| Segment       | The market segment to which the customer belongs (e.g., Consum Corporate, Home Office).                        |
| Country       | The country where the customer is located (all records are for the United States).                             |
| City          | The city where the customer is located.                                                                        |
| State         | The state where the customer is located.                                                                       |
| Postal Code   | The postal code of the customer's address.                                                                     |

|              |                                                                                             |
|--------------|---------------------------------------------------------------------------------------------|
| Region       | The geographic region of the customer (e.g., Central, East, South, West).                   |
| Product ID   | A unique identifier for each product.                                                       |
| Category     | The high-level product category (e.g., Furniture, Office Supplies, Technology).             |
| Sub-Category | A more specific classification within a product category (e.g., Bookcases, Chairs, Phones). |
| Product Name | The full descriptive name of the product.                                                   |
| Sales        | The total sales revenue generated for the product line item.                                |
| Quantity     | The number of units of the product sold in the transaction.                                 |
| Discount     | The percentage discount applied to the product for that sale.                               |
| Profit       | The profit (positive value) or loss (negative value) generated from the sale.               |

This rich dataset provides an excellent foundation for applying the analytical tools learned in CS-103, allowing for a multi-faceted exploration of the store's sales and profitability drivers.

### 3. Documentation

#### (i) Meeting Log- 1

- Date: 4/18/2026, Saturday
- Attendance: All present
- Key Discussions: The project was kicked off with an online meeting which was attended by all members. Each member was assigned with two tasks; to find a pre-existing comprehensive dataset which was to be worked on and to prepare a rough draft of a road map to follow in accordance with the project guidelines. Following the meeting the members worked on finding suitable datasets and after the input of all members and a discussion held via a WhatsApp group the "Sample Superstore" dataset was selected.

#### (ii) Meeting Log- 2

- Date: 4/20/2026, Monday
- Attendance: All present
- Key Discussions: The second meeting was held online after the members had worked on a roadmap to be followed throughout the project. The project objective was discussed as well. The members gave their input on this and agreed on a plan after which the roles were assigned.

#### (iii) Data Cleaning and Table Preparation

- Loaded the data to Power BI
- Cleaned the data, fixed any incorrect data types and checked for any missing values
- Duplicated the original superstore data to make 6 tables (Orders, Customers, Products, Geography, Shipping and Date)
- Removed unnecessary columns in each table
- Removed duplicates
- Renamed the tables to fit what they entail
- Close and Apply

#### (iv) Data Modelling

- Identified the fact and dimension tables
- Removed descriptive columns from facts tables
- Established one to many relationships between (from) dimensions to fact tables
- Set order\_date as active relationship between date table
- Set ship date as inactive relationship
- Created hierarchies for Geography and Data
- Verified all the relationships and tested the model

#### Table descriptions:

- **Orders Table:** This is the Fact Table. It contains the quantitative data for every transaction, such as Sales, Profit, and Quantity. It acts as the "hub" that connects all other tables.
- **Product Table:** A dimension table containing descriptive details about what was sold, including Category, Sub-Category, and Product Name.
- **Customers Table:** Stores specific information about the buyer, such as Customer Name and Segment (e.g., Consumer, Corporate, or Home Office).
- **Geography Table:** Contains all location-based data, including City, State, Postal Code, and Region.
- **Shipping Table:** Details how orders were delivered, primarily focusing on Ship Mode (e.g., First Class, Standard Class) and potentially Ship Dates.
- **Date Table:** A specialized table used for Time Intelligence. It allows us to analyze trends like "Sales by Month" or "Year-over-Year Growth".



## (v) SQL Queries

The foundation of the dashboard was built using MySQL, where we first designed a normalized database (`superstore_db`) with separate tables for products, customers, geography, shipping, date dimensions, and orders—ensuring referential integrity through foreign keys. After importing the raw CSV data via MySQL Workbench, we created a custom analysis table: `monthly_targets` (to store regional sales/profit targets per month) . Using this table, we wrote several analytical SQL queries to extract business insights: comparing actual sales against regional targets, analyzing customer segment performance (profit margins, average order value), identifying the most profitable shipping modes (by profit margin and shipping days), and pinpointing bottom-10 products by profit (with flags for loss makers). These queries directly informed the design of the dashboard's KPIs and visuals, such as the “Underperforming Product Sub-Categories” list and the regional performance map.

## (vi) Building the Dashboard

The dashboard follows a clean, grid-based layout with consistent spacing and a neutral background to highlight data.

- Top section – Three slicers (Year, Region) plus four KPI cards in a row.
- Left column (top) – “Monthly Targets and Actual Sales by Region” (line and stacked column chart with region as legend, month on x-axis).
- Middle column (top) – “Top 10 Products by Sub-Category” (sorted by funnel).
- Right column (top) – “Customer Segment Analysis” (clustered bar chart) “Shipme.
- Left column (bottom) - “Shipment Mode Analysis” (donut).
- Middle column (bottom) – “Profit Analysis by State” (azure map).
- Bottom right – “Underperforming Product Sub-Categories”(funnel).

All visuals share a consistent colour palette:

- Blue for actual sales / positive profit
- Orange for targets / negative profit
- Neutral grays for axes and gridlines

Three slicers were added at the top:

- Year - allows the filtering of KPI cards for all and specific years).
- Year (Targets) – allows filtering by year (All, 2014, 2015, ...). Uses the date table to propagate filters to all visuals.

- Region (Tar...) – a dropdown slicer for Central, East, West, South (region field from Monthly Targets and Actual Sales by Region table)

#### **Four KPI cards:**

The KPI cards show the total sales (revenue), total profit, average profit ratio (how much profit the business earns on average compared to sales and the average discount for the time period specified by the slicers.

#### **“Monthly Targets and Actual Sales by Region” chart:**

This chart shows the target and actual monthly sales for all 4 regions to create a comparative visual answering the question of whether the business is on budget or not.

#### **“Top 10 Products by Sub-Category” chart:**

This chart shows the top 10 best selling product sub-categories with the highest profit values.

#### **“Customer Segment Analysis” chart:**

This chart shows the profit margins for the 3 customer segments.

#### **“Shipment Mode Analysis” chart:**

This chart provides the stats to analyse the various shipment modes by giving the profit margin and order count of each.

#### **“Profit Analysis by State” chart:**

This chart (map) was needed to analyze the states using their order count and profit margin percentage.

#### **“Underperforming Product Sub-Categories” chart:**

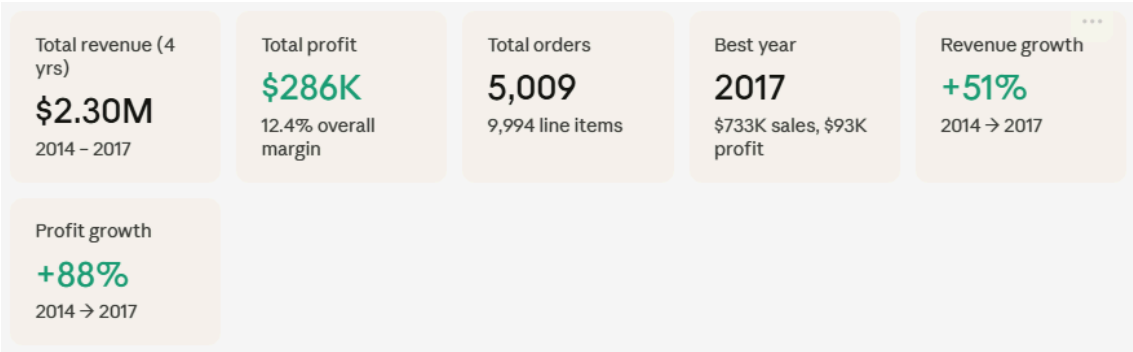
This chart highlights the underperforming product subcategories, showing the negative profit (loss) values.

#### (vii) Creating the Presentation:

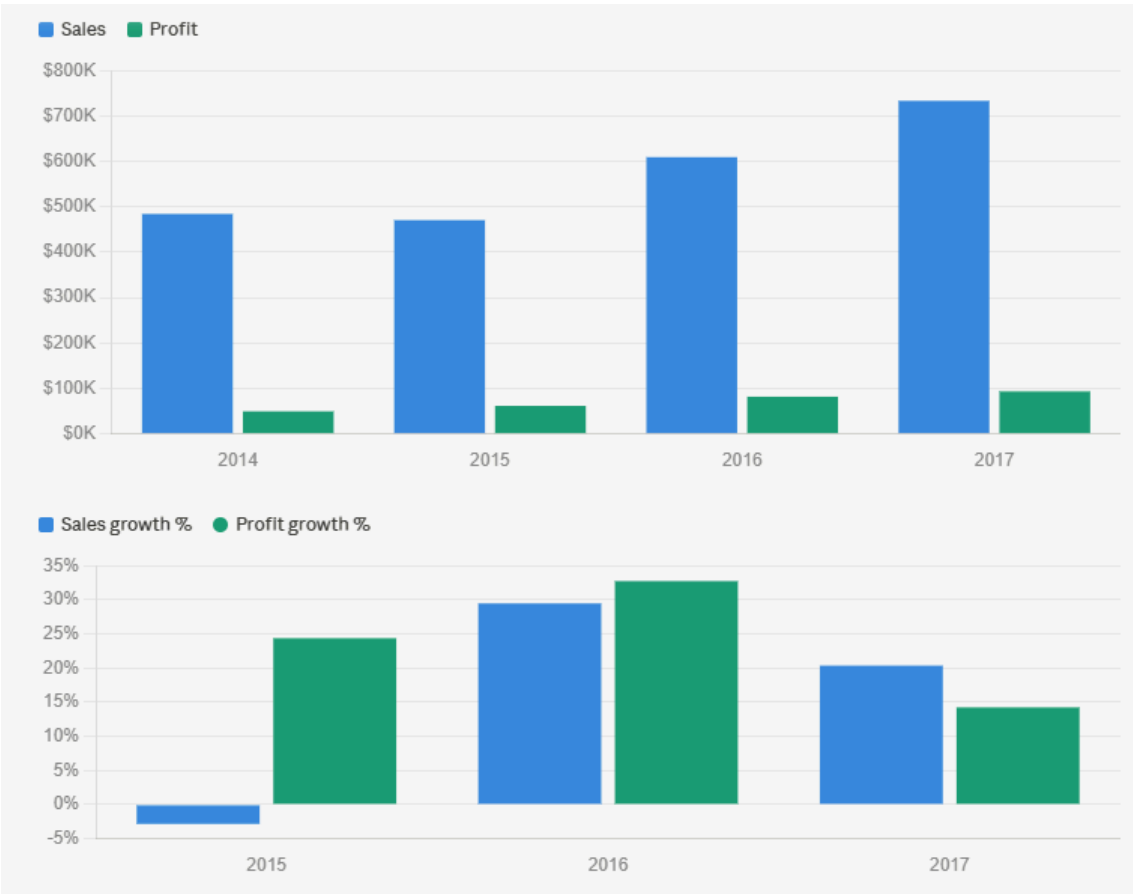
In the presentation, we walk through the entire journey—starting with sourcing the Superstore dataset from public repositories, then using power query for data cleaning and modelling, using SQL to query, filter, and aggregate the data (e.g., calculating monthly sales, profit by region, and identifying unprofitable sub-categories). After extracting the needed tables, we performed additional data cleaning and transformation in Power BI, built measures for KPIs like profit ratio and discount, designed interactive visuals (monthly trends, segment chart, shipping donut, state map, underperforming products), and assembled the final dashboard with slicers and cross-filtering. The presentation concludes with key business insights and recommendations derived from the analysis.

4. Business Insights and Recommendations Report

(Note: AI generated visuals are used in this part with the teacher’s consent.)

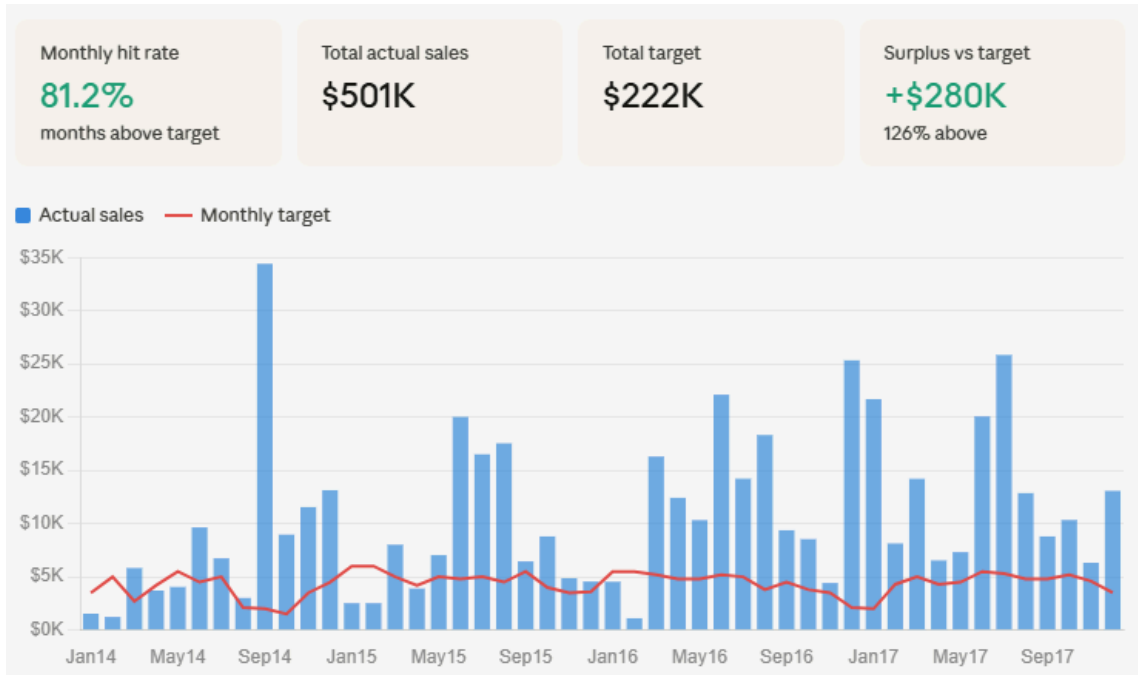


Yearly trends

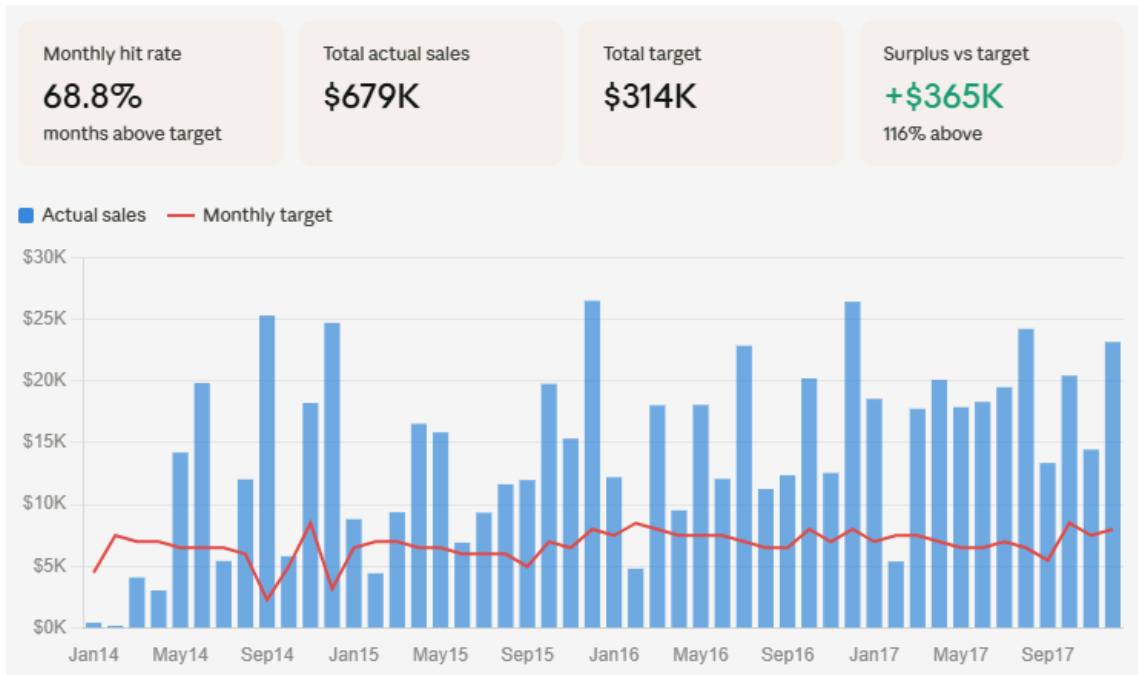


Regional targets

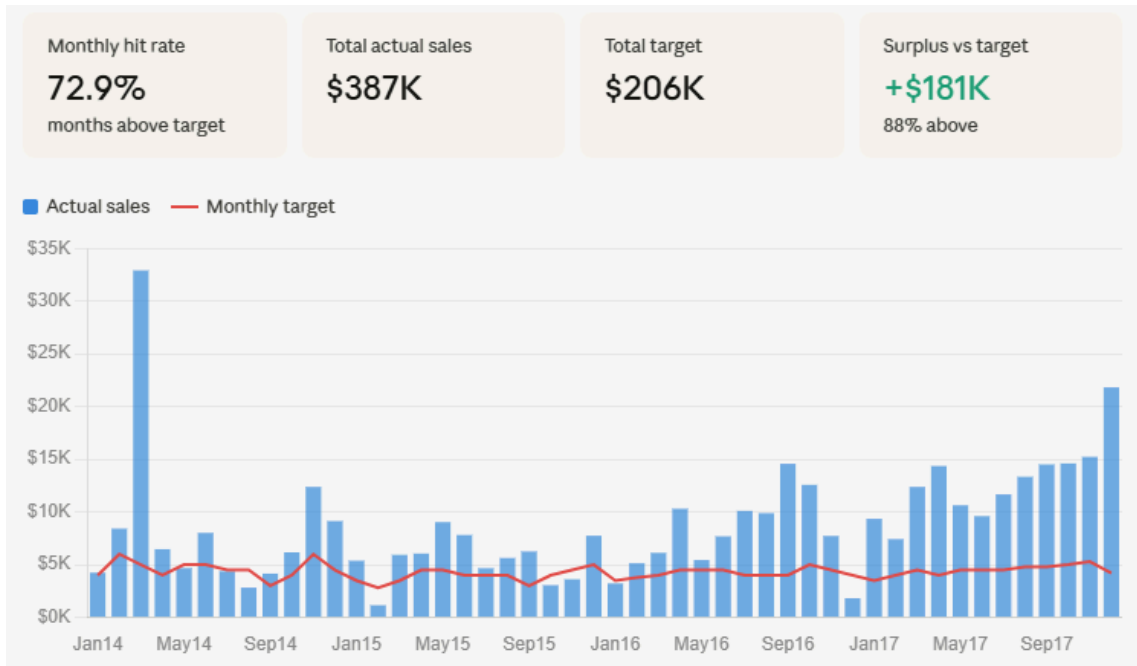
Central



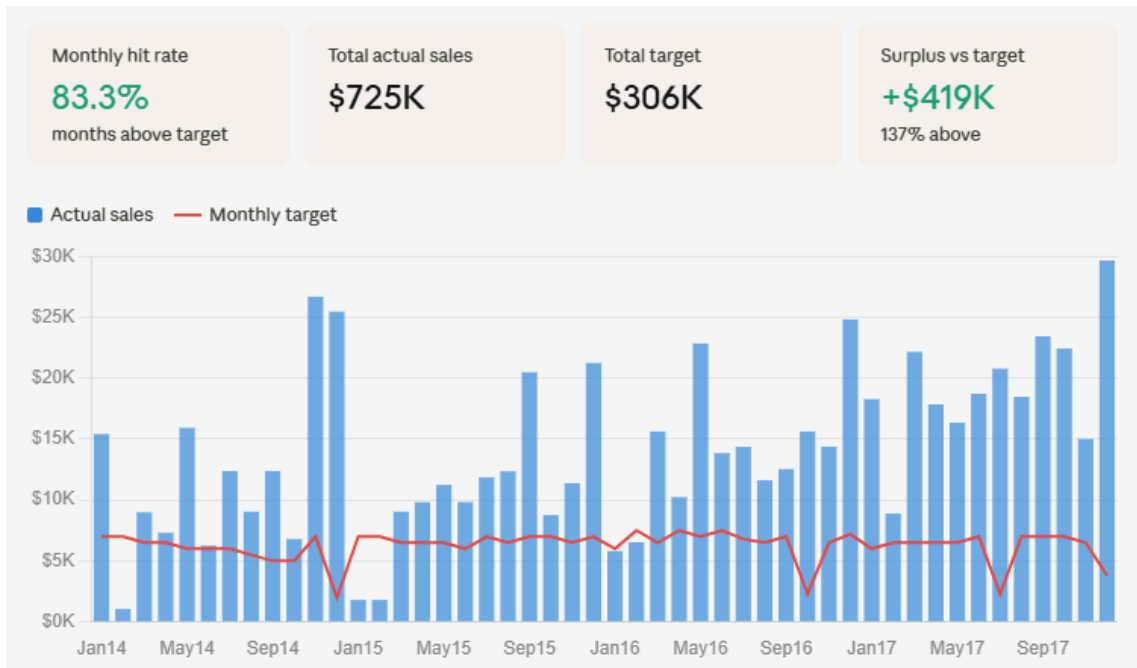
East

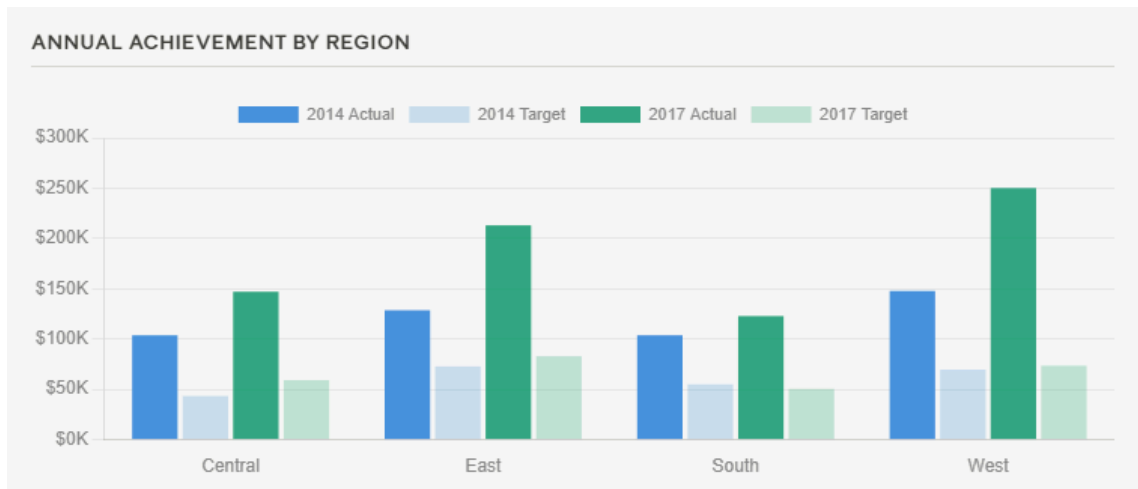


South

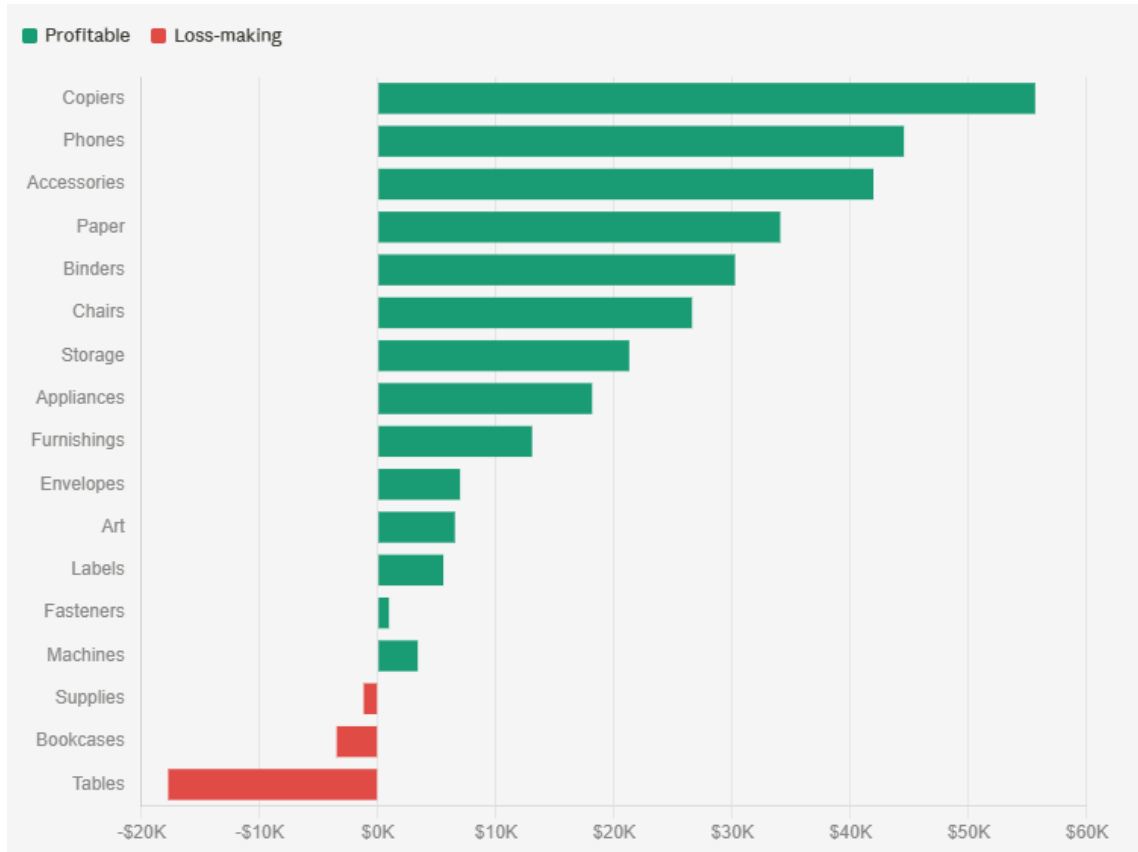


West

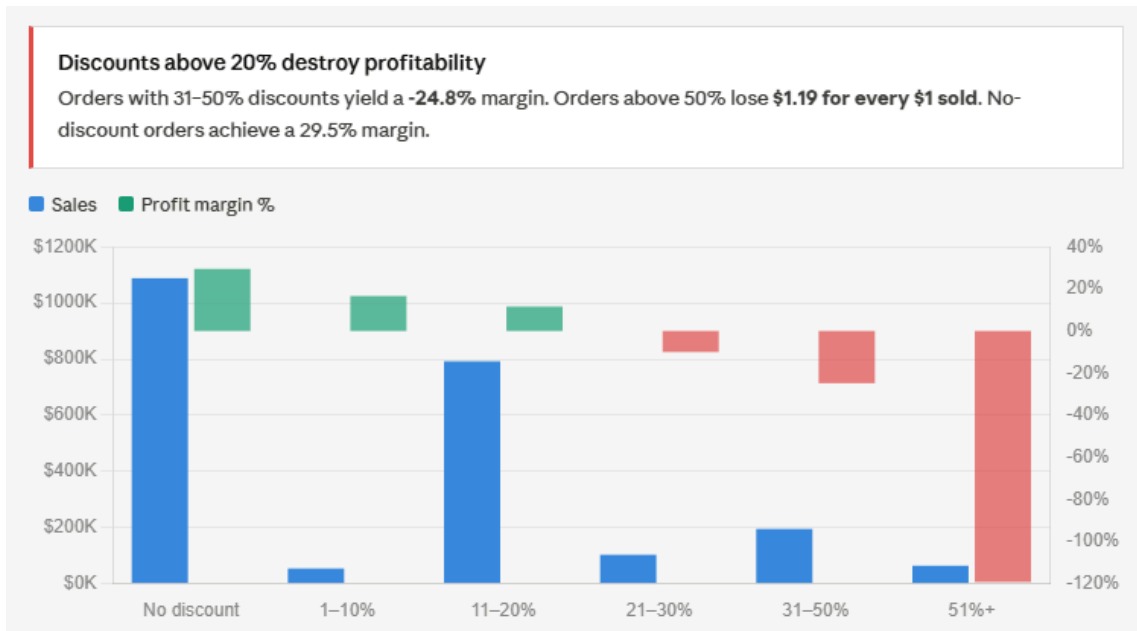




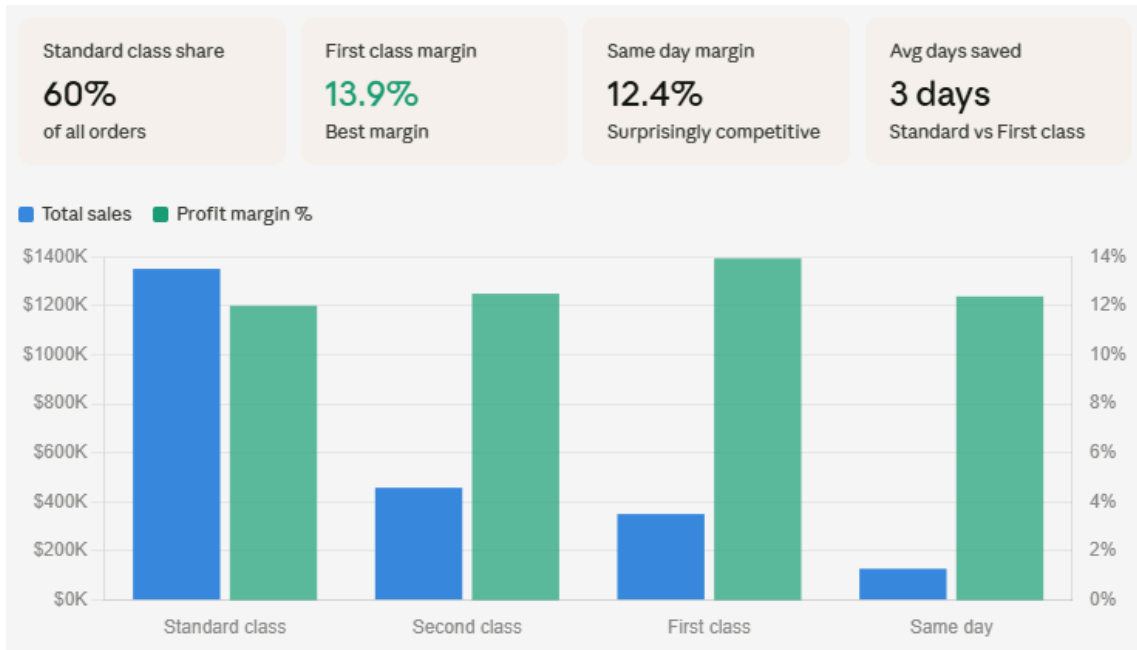
### Products



Discount impact



Shipping





## Key Insights

### **Growth** Strong multi-year revenue trajectory

Sales grew from \$484K (2014) to \$733K (2017), a 51% increase. Profit grew even faster — 88% over the same period — meaning operational efficiency improved year-on-year. 2015 was the only year with a sales dip (-2.8%), yet profit still rose 24%, suggesting cost controls kicked in.

### **Targets** All regions beat their sales targets substantially

With the corrected targets ( $\pm 10$ ), every region exceeded annual targets significantly: West over-achieved by 137%, East by 116%, Central by 127%, South by 88%. The West also achieved 100% monthly target hit-rate in 2017. This suggests targets may need upward revision to remain meaningful KPIs.

### **Seasonality** February is consistently the weakest month

Across all 4 regions and all 4 years, February accounts for the 7 worst monthly performances relative to target. East Feb 2014 achieved just 2.7% of target (\$200 actual vs \$7,500 target). This consistent pattern points to a genuine seasonal slump — marketing spend or promotions in Feb could be a growth lever.

### **Losses** Tables and Bookcases are destroying furniture profits

Tables lose \$17,725 on \$207K in sales (-8.6% margin). Bookcases lose \$3,473 (-3%). Combined, they represent \$21K in losses. In contrast, Chairs — also furniture — earn \$26K profit. The likely culprit: excessive discounting on Tables and Bookcases. These SKUs should be reviewed for minimum price floors.

### **Geography** Texas and Ohio are major profit drains

Texas generates \$170K in sales but -\$25.7K profit (-15.1% margin) — the worst in absolute dollar loss. Ohio loses \$17K despite \$78K in sales (-21.7%). Pennsylvania loses \$15.6K. All three likely have high discount rates. California and New York, by contrast, contribute \$76K and \$74K in profit respectively and should be used as benchmarks for pricing discipline.

### **Products** Technology is the profit engine

Copiers alone contribute \$55.6K profit at 37.2% margin on just 68 orders — the highest return per transaction in the business. Phones and Accessories add another \$86K combined. Technology sub-categories account for the top 3 most profitable items. Expanding the Technology sales push (particularly Copiers) could significantly lift overall margins.

### **Operations** Home Office segment punches above its weight

Home Office has the fewest orders (1,783) and lowest revenue (\$430K) but the highest profit margin at 14.0%, compared to Consumer at 11.5% and Corporate at 13.0%. This segment likely orders higher-margin Technology products with fewer discounts. It represents an opportunity to grow a high-value customer base.

## Major Findings

**Overall performance** — The store grew from \$484K to \$733K in revenue (+51%) and from \$50K to \$93K in profit (+88%) between 2014 and 2017. Profit grew faster than revenue, meaning the business became more efficient over time. The only blip was 2015, where sales dipped slightly but profit still rose 24% — strong cost control that year.

**Regional targets vs actual** — With the corrected targets ( $\div 10$ ), every single region massively over-achieved. West had the best hit rate (83.3%) and Central was close behind (81.2%). The aggregate surplus across all regions is over \$1.2M above target across the 4 years. This almost certainly means the targets need to be recalibrated upward — they're not stretching enough to be meaningful KPIs. The interactive chart lets you explore each region month by month.

**February problem** — Every region, every year, February is consistently the worst month. East Feb 2014 achieved just 2.7% of target. This is a clear seasonal pattern and a potential marketing opportunity.

**Discount crisis** — This is the most urgent operational issue. Orders with discounts above 30% are losing money — the 51%+ band loses \$1.19 for every \$1 of sales. These 1,166 high-discount orders erase roughly \$125K in profit that healthy orders generate. Capping discounts at 20% would likely unlock \$100K+ in incremental annual profit.

**Product losses** — Tables (-\$17.7K) and Bookcases (-\$3.5K) are loss-making. Copiers are the star: \$55.6K profit on only 68 orders (37.2% margin). Expanding Copier sales is likely the highest-ROI product move available.

**State-level losses** — Texas, Ohio, and Pennsylvania together lose over \$58K in profit on \$365K of sales. These are likely the same states where heavy discounting is happening and should be investigated.

## 5. Team Contributions

| Task                                                                                                                                                                                                                                         | Credits                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Finalizing the dataset and drafting the project plan                                                                                                                                                                                         | All members                                                                                                                   |
| Data cleaning, normalization and modelling                                                                                                                                                                                                   | Yasir Baig<br>Muhammad Abdullah                                                                                               |
| SQL queries and making of the targets table                                                                                                                                                                                                  | Maryam Riaz                                                                                                                   |
| Designing the Dashboard                                                                                                                                                                                                                      | Emaan Shahid                                                                                                                  |
| Project Report <ul style="list-style-type: none"><li>- Meeting logs</li><li>- Project objective, Dataset details, tables description</li><li>- Business Insights and Recommendation Report</li><li>- Appendix</li><li>- Formatting</li></ul> | All members<br>Emaan Shahid (team lead)<br>Muhammad Abdullah<br><br>Emaan Shahid<br>Yasir Baig<br>Maryam Riaz<br>Emaan Shahid |
| Presentation Making                                                                                                                                                                                                                          | Yasir Baig                                                                                                                    |
| Submission                                                                                                                                                                                                                                   | Emaan Shahid                                                                                                                  |

## 6. Team Performance Feedback Report

**Author: Team Lead**

“The team demonstrated strong collaboration throughout the project. All members attended every scheduled meeting and consistently submitted their assigned tasks on time. Beyond meeting basic expectations, the team actively supported one another—offering help whenever challenges arose and sharing input openly, which fostered a constructive and inclusive environment.

Notable strengths:

- Reliability: High adherence to deadlines and meeting attendance.
- Collaboration: Proactive peer assistance and open exchange of ideas.

- Accountability: Each member took ownership of their deliverables without needing reminders.

Overall assessment: The team's discipline and mutual support were key drivers of our on-time delivery and quality outcomes. I commend the group for setting a high standard of teamwork.

### **Challenges faced:**

The sudden shift from f2f to online teaching made it difficult to communicate with the team and the loadshedding routine made it hard to complete the tasks. However, the team managed well and the teacher also accommodated the project completion by extending the deadline.”

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### **7. Note for the teacher**

*All the supporting dataset is submitted in the zip file. In the powerBI document the data source pathways need to be changed according to their path on the teacher's device. In the sql file some tables were imported to facilitate the queries. The teacher should do the same to run the queries. Some tables were exported from the sql file in cvs and imported to powerBI via get data for the dashboard.*